

VIP C2 PROGRAMME — FULL STACK DEVELOPMENT WITH
MERN

Define Problem Statements

Online Complaint Registration and Management Systems — MERN
Stack E-commerce Application

Phase: Brainstorming & Ideation Phase

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Define Problem Statements

1. Introduction

This document formalises the problem definition artefacts for the Online Complaint Registration and Management System (OCRMS). It captures the key challenges faced by citizens, complaint resolution officers, and administrators during complaint registration, tracking, communication, and resolution processes. The identified problem statements provide the foundation for developing an efficient, transparent, and user-centric complaint management solution

2. Problem Statements

PS-1: The Online Shopper

Field	Description
I am	A complaint management officer responsible for reviewing, assigning, and resolving complaints efficiently
I'm trying to	Manage complaint records, monitor resolution progress, assign agents, and ensure timely responses to users
But	The process is highly time-consuming because complaints arrive from multiple channels and are difficult to categorize and monitor effectively
Because	Current systems lack centralized management, automated classification, analytics dashboards, and real-time communication capabilities
Which makes me feel	Overwhelmed, inefficient, and concerned about service quality and resolution performance

Impact Severity:  High —Administrative inefficiencies lead to delayed resolutions, increased workload, and reduced operational effectiveness.

PS-2: The Store Administrator

Field	Description
I am	A complaint resolution officer responsible for reviewing, investigating, and resolving complaints submitted by users
I'm trying to	Prioritize complaints efficiently, communicate with users, track resolution progress, and ensure issues are resolved within the expected timeframe
But	Complaints often arrive without proper categorization, supporting documents, or priority levels, making it difficult to determine urgency and assign them appropriately
Because	Existing systems lack automated complaint classification, centralized complaint records, real-time communication tools, and effective workload distribution mechanisms
Which makes me feel	Stressed, overloaded, and concerned that important complaints may be delayed or overlooked during the resolution process

Impact Severity: ● High — Delayed complaint resolution negatively affects user satisfaction, organizational reputation, and overall service quality.

3. Root-Cause Analysis (Ishikawa Diagram)

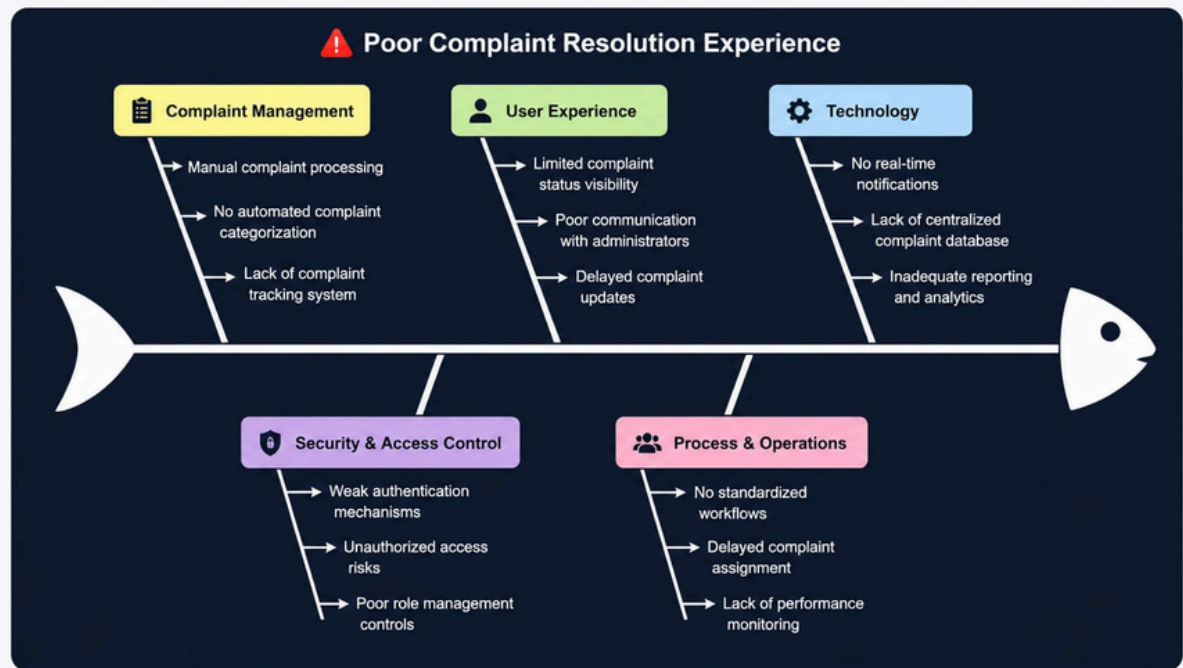


FIGURE 2.1 — ISHIKAWA (FISHBONE) ROOT-CAUSE DIAGRAM

4. Problem-to-Feature Mapping

Root Cause	Proposed Solution Feature
Manual complaint processing	Automated complaint registration and management system
Manual complaint processing	Real-time complaint status tracking dashboard
Poor communication	Integrated chat and notification system
Delayed complaint assignment	AI-based complaint classification and routing
No centralized records	MongoDB centralized complaint database
Weak authentication	JWT-based authentication and authorization
Unauthorized access risks	Role-Based Access Control (Admin, Agent, User)
Lack of analytics	Administrative dashboard with complaint reports
No notifications	Email and system notification integration

5. Conclusion

The two problem statements – representing the complaint submitter and the complaint administrator – collectively define the dual focus of the Online Complaint Registration and Management System. All identified root causes have corresponding solution features mapped within the proposed system architecture and requirement analysis documents. The proposed solution improves transparency, accountability, communication, and overall efficiency throughout the complaint resolution lifecycle.